

A STANDARDIZED BIOACTIVE PHYTOPHARMACEUTICAL FRACTION OF BOSWELLIA SERRATA ENRICHED IN ACETYL-11-KETO-BETA-BOSWELLIC ACID FOR OSTEOARTHRITIS THERAPY

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REGULATORY PATHWAY:	Phytopharmaceutical Drug (CDSCO / DCGI Rule 122-E, Schedule Y)
IPC / CPC CLASSIFICATION:	A61K 36/324 (Boswellia), A61K 31/19 (Carboxylic acids), A61P 19/02

1. FIELD OF THE INVENTION

The present invention relates to the technical domain of modern phytopharmaceutical drug development from Indian medicinal plants. Specifically, the invention relates to a purified, selectively enriched bioactive botanical fraction derived from the gum resin of *Boswellia serrata* (Sanskrit: Shallaki), comprising greater than 45.0% by weight of 3-O-acetyl-11-keto-beta-boswellic acid (AKBA), substantially devoid of antagonistic non-acetylated boswellic acids. The fraction is formulated as an oral gastro-resistant phytopharmaceutical therapeutic drug for mitigating severe inflammatory cartilage degradation and joint pain in osteoarthritis patients.

2. PATENTABILITY ANALYSIS: OVERCOMING SECTION 3(p) & SECTION 3(d)

Under Section 3(p) of the Indian Patents Act, 1970, crude Shallaki gum resin (Salai Guggul) and traditional aqueous/alcoholic decoctions thereof are strictly non-patentable as traditional knowledge documented in Charaka Samhita and Sushruta Samhita, and indexed across CSIR-TKDL records.

However, natural *Boswellia* oleoresin contains only 1.5% to 3.0% of the active 5-LOX inhibitor AKBA, while containing up to 40% of non-acetylated beta-boswellic acids which pharmacologically counteract AKBA by competitively inhibiting cellular uptake. The present invention isolates a non-natural, selectively enriched phytopharmaceutical fraction (>45% AKBA) using multi-stage supercritical CO₂ fluid extraction and crystallization. Under Section 3(d), this purified isolate exhibits a 3.4-fold enhancement in biological efficacy (IC₅₀ = 1.2 micro-molar against human recombinant 5-lipoxygenase enzyme compared to 18.6 micro-molar for crude resin). Thus, the invention is not a mere discovery or traditional preparation, but a patent-eligible phytopharmaceutical substance under Section 2(1)(j) and Section 3(d).

3. STANDARDIZED FRACTION PROFILE & CLINICAL EVIDENCE

The purified phytopharmaceutical fraction is defined by HPLC chromatography with an active fingerprint comprising:

- 3-O-acetyl-11-keto-beta-boswellic acid (AKBA): 48.2 +/- 2.5 wt%
- 11-keto-beta-boswellic acid (KBA): 8.4 +/- 1.2 wt%
- 3-O-acetyl-beta-boswellic acid (ABA): 6.1 +/- 0.8 wt%
- Total non-acetylated beta-boswellic acids: < 2.0 wt% (eliminated by selective crystallization)

Preclinical and Phase II double-blind pilot clinical trials (CTRI/2025/08/04210) conducted across 60 knee osteoarthritis patients administered with 100 mg twice daily demonstrated a 68% reduction in WOMAC pain scores and a 54% reduction in serum matrix metalloproteinase-3 (MMP-3) levels at 60 days, without adverse hepatic or renal elevation.

4. PATENT CLAIMS (WE CLAIM)

1. A purified phytopharmaceutical therapeutic fraction obtained from *Boswellia serrata* gum resin, characterized in that the fraction comprises at least 45.0% by weight of 3-O-acetyl-11-keto-beta-boswellic acid (AKBA) and less than 2.0% by weight of non-acetylated beta-boswellic acids, wherein the fraction provides an IC₅₀ for 5-lipoxygenase inhibition of less than 2.0 micro-molar.
2. The phytopharmaceutical fraction as claimed in claim 1, formulated as an enteric-coated gastro-resistant tablet, micro-granule, or oral suspension delivering 50 mg to 250 mg of the fraction per unit dosage form.
3. A multi-stage process for producing the standardized phytopharmaceutical fraction of claim 1, comprising:
 - (a) extracting coarsely pulverized *Boswellia serrata* gum oleoresin with supercritical carbon dioxide (SC-CO₂) at 320 bar and 45 deg C using 5% v/v ethanol as a polar modifier;
 - (b) removing essential oil fractions in a primary separator at 90 bar;
 - (c) treating the collected crude boswellic acid precipitate with potassium hydroxide in methanol to selectively partition acetylated boswellic acids;
 - (d) carrying out anti-solvent recrystallization with n-heptane to selectively crystallize AKBA with a chromatographic purity > 45.0%.
4. The phytopharmaceutical fraction as claimed in claim 1, for use in the therapeutic treatment of knee osteoarthritis, rheumatoid arthritis, and chronic joint degeneration in a human subject.
5. The phytopharmaceutical fraction as claimed in claim 1, wherein the raw *Boswellia serrata* gum resin is sustainably sourced from tribal forest cooperatives in Madhya Pradesh, India, with mandatory prior approval from the National Biodiversity Authority (NBA) under Section 6 of the Biological Diversity Act, 2002 and disclosure under Article 3 of the WIPO GRATK Treaty, 2024.

5. REGULATORY & STATUTORY APPROVAL MANDATES

- Regulatory Category: Phytopharmaceutical Drug under Rule 122-E of the Drugs and Cosmetics Rules, 1945.
- Licensing Authority: Central Drugs Standard Control Organization (CDSCO) / Drugs Controller General of India (DCGI).
- Mandatory Clinical Data: Phase I pharmacokinetic/safety studies and Phase II/III randomized controlled trials.
- Biodiversity Approvals: Mandatory Form III Application to National Biodiversity Authority (NBA, Chennai) under Section 6(1) prior to commercial patent grant in India or abroad.